

Justin Nitoi

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EDUCATION

University of Waterloo

Sep 2024 — Dec 2028

Bachelor of Computer Science

Waterloo, Ontario

- **GPA: 4.0/4.0** | Dean's Honours List | President's Scholarship of Distinction.
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Calculus II, Probability & Statistics, Logic & Computation.
- **Awards:** Euclid Math Contest Honour Roll (Top 1% in Canada), AIME Qualifier.

EXPERIENCE

AI Automation Developer (8 months)

May 2026 — Dec 2026

CGI Group Inc - Systems Integration & Consulting

Toronto, Ontario

- Engineered an enterprise RAG microservice using **FastAPI**, **LangChain**, and **Azure OpenAI**, automating IT support ticket triage and reducing manual processing time by **65%**.
- Architected an asynchronous data ingestion pipeline leveraging **Celery**, **Redis**, and **Azure Blob Storage** to process and embed **100,000+** enterprise documents into a **PostgreSQL (pgvector)** database.
- Optimized vector search queries and implemented dynamic API caching, achieving a **P99 retrieval latency of under 150ms** while supporting up to **5,000+ daily concurrent requests**.
- Containerized AI workloads with **Docker** and orchestrated CI/CD pipelines via **GitHub Actions**, improving deployment reliability and cutting environment setup time by **40%**.

AI Engineering Consultant (4 months)

Jun 2025 — Aug 2025

Story City

Remote

- Architected **3 AI microservices** using **Azure** and **Google Maps API**, engineering a graph algorithm that reduced storyboard creation time by **40%**.
- Optimized route duration prediction logic to within **± 5 minutes** and migrated infrastructure to **AWS Lambda**, cutting projected cloud costs by **50%**.
- Presented technical feasibility reports to stakeholders, directly influencing the roadmap for **2 core product features**.

PROJECTS

Stremio Movie Catalog Microservice | Python (Flask), Redis

Feb 2026 — Present

- Engineered a high-throughput **RESTful API** serving custom movie catalogs to over **1,000+ active users** on Stremio, handling peak traffic without downtime on Render.
- Implemented a **Redis caching layer** to store TMDB API responses, reducing upstream latency by **90%** and preventing external rate-limiting bottlenecks.
- Designed efficient JSON serialization pipelines to parse external movie metadata, optimizing response times to under **100ms** for end-users.

NameNet (Transformer Research) | Python, PyTorch

Jun 2025 — Aug 2025

- Engineered a gender classification model using **PyTorch**, benchmarking LSTM vs. Transformer architectures on **40k+** samples.
- Implemented data sanitization with **Pandas** to correct label inconsistencies, boosting validation accuracy from 88% to **99.2%**.

SKILLS

- **Programming Languages:** Python, C++, SQL, JavaScript/Typescript, HTML/CSS, Java
- **Technologies:** PyTorch, Keras, TensorFlow, scikit-learn, Pandas, FastAPI, Flask, Node.js, React.js, Git/GitHub, Linux, Docker, AWS Lambda, Azure Functions, JIRA, VS Code, PostgreSQL
- **Certifications:** Microsoft Certified: Azure Fundamentals, Azure AI Fundamentals